

Quantifying the Health Benefits Linked to Reduced Traffic-Related PM_{2.5} Exposure on Acute Coronary Syndrome Incidence in China

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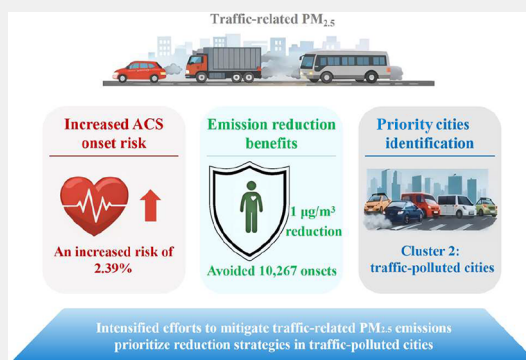
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ABSTRACT: The transportation sector powered by fossil fuels is a significant contributor to PM_{2.5} pollution. Systematic evaluation of traffic-related PM_{2.5} on acute coronary syndrome (ACS) onset is not yet characterized, and the health benefits of reduced traffic-related PM_{2.5} and evidence-based mitigation strategies are currently absent. We investigated the association of ACS onset with short-term traffic-related PM_{2.5} exposure in a nationwide time-stratified case-crossover study, using 627,828 ACS cases extracted from the China Cardiovascular Association. We evaluated the risks of ACS onset associated with traffic-related PM_{2.5}, calculated the attributable onsets, and then conducted cluster analysis using the K-means algorithm to identify priority cities for emission reduction. We found a robust association between traffic-related PM_{2.5} and ACS onset, i.e., an increased risk of 2.39% (95% CI: 1.79–3.00%) for same-day (lag 0) PM_{2.5} per 10 µg/m³ increase. The greatest health benefits due to a 1 µg/m³ PM_{2.5} reduction were achieved by traffic-related PM_{2.5}, reaching 10,267 (95%CI: 4503–16,063) onsets per year. In addition, three obvious categories were clustered, and traffic-polluted cities were highlighted (cluster 2, not belonging to megacities), with high emissions and concentrations of traffic-related PM_{2.5}. Therefore, intensified efforts to mitigate traffic-related PM_{2.5} emissions should be promoted, and the prioritization of emission reduction strategies in these traffic-polluted cities is imperative for safeguarding public health.

KEYWORDS: traffic-related PM_{2.5}, acute coronary syndrome, acute onset, case-crossover, health benefits



1. INTRODUCTION

Transportation is a fundamental component of cities, influencing urbanization and its development.¹ According to a United Nations' report, 55% of the world's population lived in cities in 2018, and this proportion would reach 68% by 2050, placing more stress on the public transportation system.^{2,3} The transportation sector, mainly driven by fossil fuels, consumes most of the world's liquid fuels and contributes roughly 10% to global emissions of primary PM_{2.5}.^{4,5} An increasing number of epidemiologic studies support that short-term exposure to traffic-related PM_{2.5} increases health risks, exceeding coal combustion, which is generally the most concerning source.^{6–10} The Guidance to Reduce the Cardiovascular Burden of Ambient Air Pollutants, released by *Circulation*, has proposed a special interest in traffic-related PM_{2.5}.¹¹

Currently, inconsistent results have been reported for cardiovascular outcomes as some studies have demonstrated significant associations with traffic-related PM_{2.5},^{6,7,12,13} whereas others have not found compelling evidence.^{14–16} In addition to PM_{2.5}, a variety of traffic-related factors, such as elemental and black carbon,^{17,18} gaseous pollutants,¹⁸ time spent in

traffic,^{19,20} roadway proximity,^{21,22} and traffic noise,^{23,24} have been linked to cardiovascular effects. Overall, most of the previous studies primarily focused on total cardiovascular mortality, hospital admissions, and emergency visits due to limited credible and specific disease data.

Acute coronary syndrome (ACS) is the leading cause of morbidity and mortality worldwide.^{25–28} Several recent studies have found links between short-term exposure to PM_{2.5} and ACS outcomes,^{29–32} but evidence linking PM_{2.5} and ACS remains unquantified.³³ A large-scale analysis highlighted that hourly exposure to PM_{2.5} would trigger ACS onset using a case-crossover analysis in 1.29 million Chinese patients and reported an elevated risk of ACS onset with proximity to traffic.²² To date, only limited studies have explored the acute

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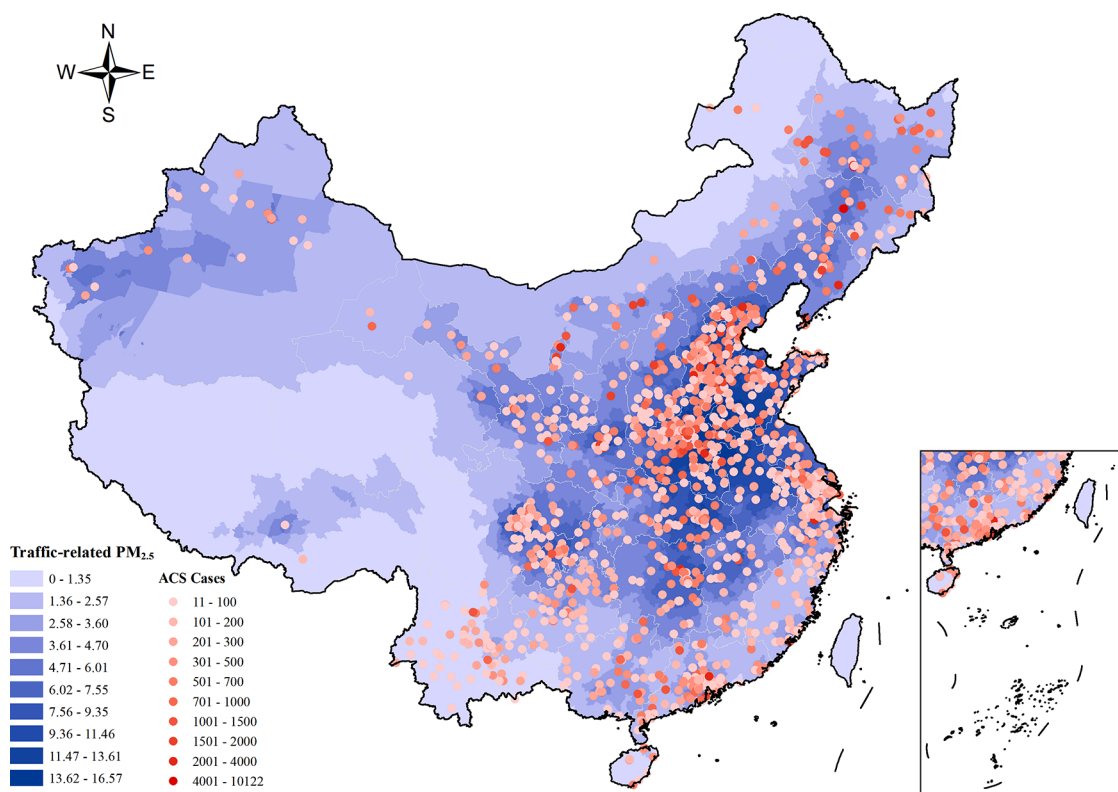


Figure 1. Spatial distributions acute coronary syndrome (ACS) cases and traffic-related PM_{2.5}, 2015 to 2018. Blue represents the annual exposure level of traffic-related PM_{2.5}, and pink dots show the locations of 1654 hospitals with total acute coronary syndrome (ACS) cases. Different gradations of colors show different levels of exposure and outcome.

effects of traffic-related PM_{2.5} on myocardial infarction (MI), which is a subtype of ACS. For example, a previous study in California reported significant correlations between PM_{2.5} from vehicular emissions and MI emergency department visits.¹² Notably, the association of traffic-related PM_{2.5} with ACS onset is not yet fully characterized, and comprehensive, large-scale population studies are lacking. Therefore, precise measures for targeted emission reduction and health protection in key areas appear to be inadequate.

China is the world's largest producer and consumer of fossil fuel energy, accounting for 21% of global energy consumption and leading to severe PM_{2.5} pollution.³⁴ The Chinese government has released a package of policies on air pollution control through emission reduction (particularly those from coal combustion) and renewable energy promotion, achieved a 58.3% decrease in PM_{2.5} concentrations between 2013 and 2023.^{35,36} However, traffic-related air pollution, along with the rapid urbanization process and vehicle growth, is still serious, compounded by the scarcity of evidence-based health impacts of traffic-related PM_{2.5} and the lack of established protocols for identifying priority mitigation areas.³⁷ Meanwhile, China has constructed the largest nationwide rescue network, the National Chest Pain Center, for patients with acute chest pain, recently attracting 2096 accredited hospitals and covering more than 96% of counties in China. It has acquired massive records for ACS and other acute cardiovascular diseases from prehospital and in-hospital care, enabling individual-level analysis for specific diseases.^{28,38} This provides a unique opportunity to investigate the role of traffic-related PM_{2.5} in ACS onset and assess additional onsets in China.

Herein, ACS onset records from the National Chest Pain Center from 2015 to 2018 and traffic-related PM_{2.5} from the source-oriented Community Multiscale Air Quality (CMAQ) model were used to analyze the association of ACS onset with short-term exposure to traffic-related PM_{2.5} in a time-stratified case-crossover study. This study aims to (1) demonstrate the short-term impact of traffic-related PM_{2.5} on ACS onset and identify populations at risk; (2) estimate additional ACS onsets attributable to traffic-related PM_{2.5} and examine potential health benefits from concentration reductions of traffic-related PM_{2.5}; and (3) identify cities that should prioritize traffic-related PM_{2.5} reductions and propose targeted policy recommendations.

2. METHODS

2.1. Data Collection

We extracted ACS onset records from the CCA Database Chest Pain Center, between January 1, 2015, and December 31, 2018, including ST-elevation myocardial infarction (STEMI), International Classification of Diseases, 10th Revision (ICD-10): I21.0-I21.3; non-ST-elevation myocardial infarction (NSTEMI), ICD-10 code: I21.403; and unstable angina (UA), ICD-10 code: I20.001. ACS diagnoses were determined using strict and uniform standards in the National Chest Pain Center network, based on comprehensive judgment of past medical history, symptoms, and signs, combined with electrocardiogram and laboratory examinations. Details of the ACS diagnosis are presented in the [Supporting Information \(Method S1\)](#). In order to cover all provincial-level regions, 1654 certified hospitals from 1155 counties were selected, with special emphasis on hospitals approved by the regional certification center and with ACS records exceeding the 20th percentile (i.e., 10 individuals) ([Figure 1](#)). Simultaneously, we collected data on demographic characteristics (e.g., age, sex, race,

and marital status) and onset indicators (e.g., onset date, emergency, or hospital admission ID). This study was approved by the Ethical Committee of the National Institute of Environmental Health, Chinese Center for Disease Control and Prevention (No. 202102).

2.2. Exposure Assessment

We conducted the source-oriented Community Multiscale Air Quality (CMAQ) model (version 5.2) with a $36 \times 36 \text{ km}^2$ resolution to generate daily concentrations of $\text{PM}_{2.5}$ sources from 2015 to 2018 in China. The model description and performance had been described in previous studies.^{39–41} Then, the data were downscaled at a spatial resolution of $0.1^\circ \times 0.1^\circ$ (approximately $10 \times 10 \text{ km}^2$) using inverse distance weighted (IDW) interpolation^{42,43} and adjusted by integrating the database of Tracking Air Pollution in China (TAP).⁴⁴ The TAP database was developed and maintained by Tsinghua University. Out-of-bag cross-validation (CV) was performed to test the fitting of TAP with ground observations, which showed good agreements, with an R^2 of 0.80–0.88 between 2013 and 2020.⁴⁵ Details of model validation, adjustment, and downscaling are shown in the Supporting Information (Method S2). Daily concentrations of traffic-related $\text{PM}_{2.5}$ in each county were calculated as the average of all of the filled grids (including partly). In addition, daily exposure data were matched based on the hospital name of ACS patients and the district or county in which the hospital was located.

2.3. Statistical Analyses

2.3.1. Exposure–Response Relationship Assessment. We performed a time-stratified case-crossover analysis and conditional logistic regression models to determine the associations of traffic-related $\text{PM}_{2.5}$ with ACS onset, using the odds ratio (OR) to represent the health risk for a $10 \text{ }\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ exposure. We stratified the model by each ACS onset case and assigned each case as “1” and each control as “0”. Consistent with previous studies, a natural cubic spline with 3 degrees of freedom (df) was used to control the daily temperature, with relative humidity as a linear covariate and public holidays as a binary variable.⁴⁶ Single and moving average lag effects were investigated, with the maximum lag set to 3 days. The same-day (lag 0), with the most significant effect estimate, was determined as the primary analysis. Moreover, associations among subtypes (STEMI, NSTEMI, and UA) and subgroups were demonstrated, including age (<65 vs ≥ 65 years), sex (male vs female), season (warm vs cold), and region (south vs north). Potential modification effects and differences between the two subgroups were examined using the Z test.^{47,48} Two-sided statistical tests were used in the stratified analysis, and $p < 0.05$ was considered statistically significant.

In addition, multiple strategies were adopted for sensitivity analyses to check the robustness of the main model. First, we checked different dfs (4, 5 and 6) and linear type for daily temperature, and a natural cubic spline with 3 df for relative humidity was checked. Second, natural cubic splines with the same dfs (i.e., 3, 4, 5, and 6) for temperature and relative humidity were simultaneously adjusted. Third, a copollutant model of ozone was performed utilizing the TAP database to examine robustness. The results of the exposure assessment at a $36 \times 36 \text{ km}^2$ resolution are also presented.

2.3.2. Additional ACS Onset Estimation. The optimal lag with the maximum effect estimate, i.e., same-day (lag 0), was used to calculate additional ACS onsets in 2848 counties on the Chinese mainland in 2015 and 2018 from traffic-related $\text{PM}_{2.5}$. The average daily hospitalizations for ACS in Beijing were utilized, drawn from the Beijing Municipal Health Big Data and Policy Research Center. In this study, $0 \text{ }\mu\text{g}/\text{m}^3$ was selected as the reference value (Method S3). In addition, avoided ACS onsets due to a $1 \text{ }\mu\text{g}/\text{m}^3$ reduction in total and source-specific sources $\text{PM}_{2.5}$ concentrations were also calculated.

2.3.3. Cluster Analysis of Different Transportation Modes. According to the “2017 Annual Analysis Report on Transportation in Major Cities in China”, 98 cities with significant representativeness were selected in this study. These cities spanned across 30 provinces (except Xizang) in China, encompassing a range of scale levels from first-tier cities to small- and medium-sized cities, ensuring diversity and comprehensiveness. Socio-economic characteristics (i.e., per

capita gross domestic product and resident population), transportation indicators (i.e., traffic congestion index, civilian vehicle ownership, and traffic-related $\text{PM}_{2.5}$ emissions and concentrations), and health indicators (i.e., additional ACS onsets) were collected and then clustered using the K-means algorithm.

We performed all data preparation and analyses using R software (version 4.0.2). For example, the “raster” package was used to generate meteorological data in the form of time series, and the “survival” package was used to run conditional logistic regression models.

3. RESULTS

3.1. Descriptive Analysis

A total of 627,828 individuals from 1654 hospitals were diagnosed with ACS onset from 2015 to 2018. Among them, 67.7% were males, 48.0% were more than 65 years, and 61.1% lived in the north of China. As for cause-specific diseases, 43.3% were diagnosed with unstable angina (UA), and 38.0% were diagnosed with ST-elevation myocardial infarction (STEMI) (Table 1). The spatial distribution of inclusion cases and traffic-related $\text{PM}_{2.5}$ exposure at the county level is depicted in Figure 1, and the southeastern half of China experienced slightly higher exposure levels simultaneously. Further subdivision of the exposure window into the case and

Table 1. Demographic Characteristics of Patients Admitted for Acute Coronary Syndrome (ACS) and Corresponding Exposure Levels on Case and Control Days^a

indicators	cases			
	ACS	STEMI	NSTEMI	UA
total	627,828	238,790	117,025	272,013
sex				
male	425,036	181,961	80,986	162,089
female	202,792	56,829	36,039	109,924
age (years)				
<65	326,389	132,450	56,258	137,681
65~74	175,278	61,841	32,696	80,741
≥ 75	126,161	44,499	28,071	53,591
season				
warm	300,493	113,357	55,043	132,093
cold	327,335	125,433	61,982	139,920
region				
south	244,136	97,370	53,348	93,418
north	383,692	141,420	63,677	178,595
on case days				
traffic-related $\text{PM}_{2.5}$ ($\mu\text{g}/\text{m}^3$)	7.33 (7.02)	7.29 (6.90)	7.27 (6.87)	7.38 (7.18)
mean temperature ($^\circ\text{C}$)	14.18 (11.35)	14.03 (11.28)	14.52 (11.08)	14.17 (11.51)
relative humidity (%)	65.94 (18.20)	66.00 (18.23)	67.20 (17.79)	65.34 (18.33)
on control days				
traffic-related $\text{PM}_{2.5}$ ($\mu\text{g}/\text{m}^3$)	7.23 (6.95)	7.19 (6.84)	7.18 (6.85)	7.28 (7.09)
mean temperature ($^\circ\text{C}$)	14.17 (11.37)	14.05 (11.28)	14.52 (11.09)	14.13 (11.57)
relative humidity (%)	65.89 (18.27)	66.07 (18.29)	67.24 (17.82)	65.15 (18.40)

^aNote: ACS, acute coronary syndrome; STEMI, ST-elevation myocardial infarction; NSTEMI, non-ST-elevation myocardial infarction; UA, unstable angina. Warm season: April to September; cold season: October to March. Mean (standard deviation, SD) values were listed on case and control days.

control days showed an estimated mean exposure to traffic-related $PM_{2.5}$ of 7.33 (7.02) $\mu g/m^3$ on case days and slightly higher than those on control days. The overall county-level exposures are presented in Table S1.

3.2. Regression Results

The estimations for ACS under multiple lag structures are shown in Figure S1. The largest magnitude of traffic-related exposure for a $10 \mu g/m^3$ increase in concentration was observed at lag 0 day (i.e., same-day). Meanwhile, traffic-related $PM_{2.5}$ was significantly associated with an elevation in ACS onset risk (OR = 1.0239, 95% CI: 1.0179–1.0300) for a $10 \mu g/m^3$ change, and it persisted among the three specific causes (Figure 2). Furthermore, traffic-related $PM_{2.5}$ had the largest associations with STEMI (OR = 1.0370, 95% CI: 1.0270–1.0470).

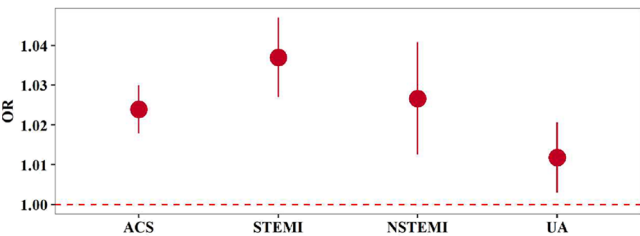


Figure 2. Short-term effects of traffic-related $PM_{2.5}$ on the onset of acute coronary syndrome (ACS) and its subtypes. Health effects are shown as odds ratios (ORs) of the onsets to traffic-related $PM_{2.5}$ with an increase of $10 \mu g/m^3$ on the same-day (lag 0); STEMI, ST-elevation myocardial infarction; NSTEMI, non-ST-elevation myocardial infarction; UA, unstable angina.

Although different dfs, copollutants (i.e., ozone), and other data sets were examined, negligible changes were observed in

the sensitivity analysis (Table S2), indicating that the estimated health risks were relatively robust. Besides, the difference was more significant for individuals aged over 65 years than for those aged under 65 years ($p = 0.049$), and females were shown to have a relatively higher risk of non-ST-elevation myocardial infarction (NSTEMI) than males (Figure 3). Stratified analyses of region and season are shown in Figure S2.

3.3. Attributable ACS Onsets

The estimated additional admissions for ACS onset due to traffic-related $PM_{2.5}$ exposures were 71,722 (95% CI: 31,348–112,627) in 2015 and 63,864 (95% CI: 27,914–100,272) in 2018 (Figure S3). UA was the most abundant component, accounting for 50% of ACS onsets due to traffic-related $PM_{2.5}$. Figure 4 depicts the spatial distributions of traffic-related $PM_{2.5}$ exposures and the corresponding ACS onsets in 2015 and 2018. The Beijing–Tianjin–Hebei region and the surrounding areas experienced relatively higher traffic-related $PM_{2.5}$ exposures. Slightly decreasing concentrations of $PM_{2.5}$ were observed in these areas in 2018. The concentrations of traffic-related $PM_{2.5}$ in 2018 increased by more than $0.5 \mu g/m^3$ in some parts of central China, Guangdong, and Jiangxi province (panel c). The spatial distribution of additional ACS onsets depended on population density and exposure levels, and no considerable tendency for spatial variations was observed. Generally, the Sichuan–Chongqing–Guizhou provincial boundary region and the Pearl River Delta region showed higher numbers of onsets (panel f).

3.4. Avoided Onsets Comparisons

Significant avoided onsets of STEMI, NSTEMI, and UA were observed due to a $1 \mu g/m^3$ reduction in total and source-specific $PM_{2.5}$ concentrations (Figure 5). There were 1600 avoided onsets due to total $PM_{2.5}$ reduction, not as significant

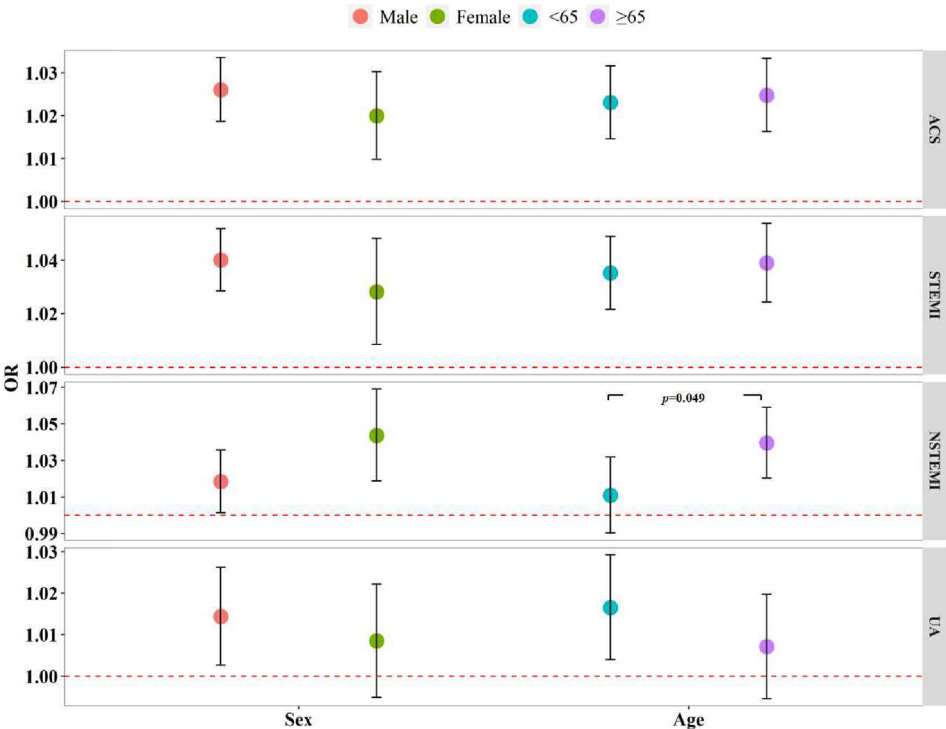


Figure 3. Odds ratios (ORs) of acute coronary syndrome (ACS) to traffic-related $PM_{2.5}$ with an increase of $10 \mu g/m^3$ in stratified analysis on the same-day (lag 0), classified by sex and age.

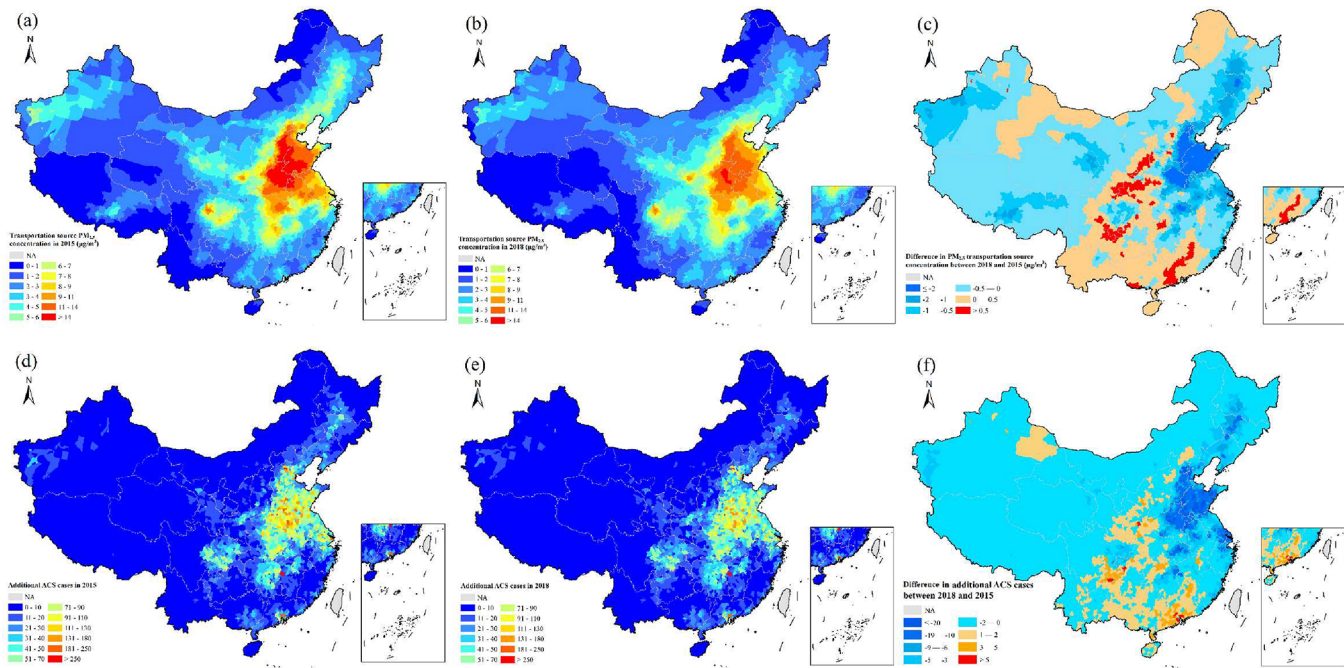


Figure 4. Spatial distribution of traffic-related PM_{2.5} exposures and corresponding acute coronary syndrome (ACS) onsets in 2015 and 2018. (a) Traffic-related PM_{2.5} concentrations in 2015; (b) traffic-related PM_{2.5} concentrations in 2018; (c) difference in concentrations between 2018 and 2015, with negative (blue) ones indicating traffic-related PM_{2.5} quality improvement and positive ones (brown to red) representing unchanged or slightly worse; (d) additional acute coronary syndrome (ACS) onsets due to traffic-related PM_{2.5} exposure in 2015; (e) additional acute coronary syndrome (ACS) onsets in 2018; (f) difference in additional onsets between 2018 and 2015, with negative (blue) ones indicating health benefits from 2015 to 2018 and positive ones (brown to red) representing increased onsets.

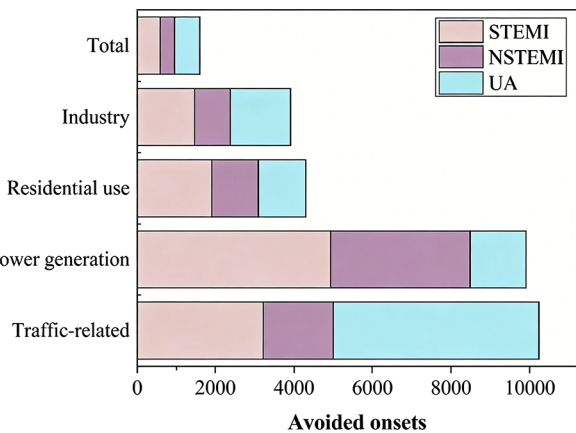


Figure 5. Avoided onsets due to a 1 µg/m³ reduction of total and source-specific PM_{2.5} concentrations.

as targeted reductions in source-specific PM_{2.5}. In contrast, traffic-related PM_{2.5} was associated with the highest number of avoided ACS onsets, i.e., 10,267 (95%CI: 4503–16,063). Therefore, it was recommended that we give priority to controlling and reducing emissions specifically from traffic-related PM_{2.5} in order to obtain the greatest health benefits.

3.5. Cluster Analysis

In this study, three obvious categories were clustered in 98 Chinese cities (Figure 6). First, cities with severe traffic-related PM_{2.5} pollution were identified (cluster 2, traffic-polluted cities), such as Zibo in Shandong, Handan in Hebei province, Xinxiang in Henan, and Xuzhou in Jiangsu. They were characterized by low per capita gross domestic product, moderate resident population, and relatively low number of

motor vehicles and traffic congestion indices. Nevertheless, such cities exhibited high emissions and concentrations of traffic-related PM_{2.5}, resulting in very high ACS onsets (Tables S3 and S4). Prioritizing emission reductions, promoting clean-energy vehicles, and delivering early health protection alerts were suggested to achieve lowering onsets and protect the public. Second, cities like Beijing and Shanghai (cluster 1, traffic-intensive cities) were densely populated urban areas with high socio-economic status, resident population, motor vehicle ownership, traffic-related PM_{2.5} emissions, and traffic congestion indices, leading to relatively high ACS onsets. It was recommended to control the growth of motor vehicles and implement restrictions to alleviate traffic congestion, achieving reductions of traffic-related PM_{2.5} emissions. The third category consisted of traffic-friendly cities (Cluster 3), with lower levels of socio-economic development and traffic-related indicators, leading to lower ACS onsets.

4. DISCUSSION

This study illustrated a robust association between traffic-related PM_{2.5} concentration and ACS onset using a nationwide case-crossover analysis of 627,828 ACS cases from 1654 hospitals in China. It persisted among three specific causes, STEMI, NSTEMI, and UA. The largest evaluation was observed with STEMI (increased by 3.70%, 95% CI: 2.70–4.70%) for every 10 µg/m³ increment in PM_{2.5} concentrations. Meanwhile, decreased ACS onsets and diversified spatial characters from 2015 to 2018 were simultaneously observed from traffic-related exposure, and higher health benefits due to per unit of reduction were also observed. Furthermore, three distinct types of cities were clustered in this study, and traffic-polluted cities (cluster 2, not belonging to megacities, with moderate resident populations but high levels of transportation

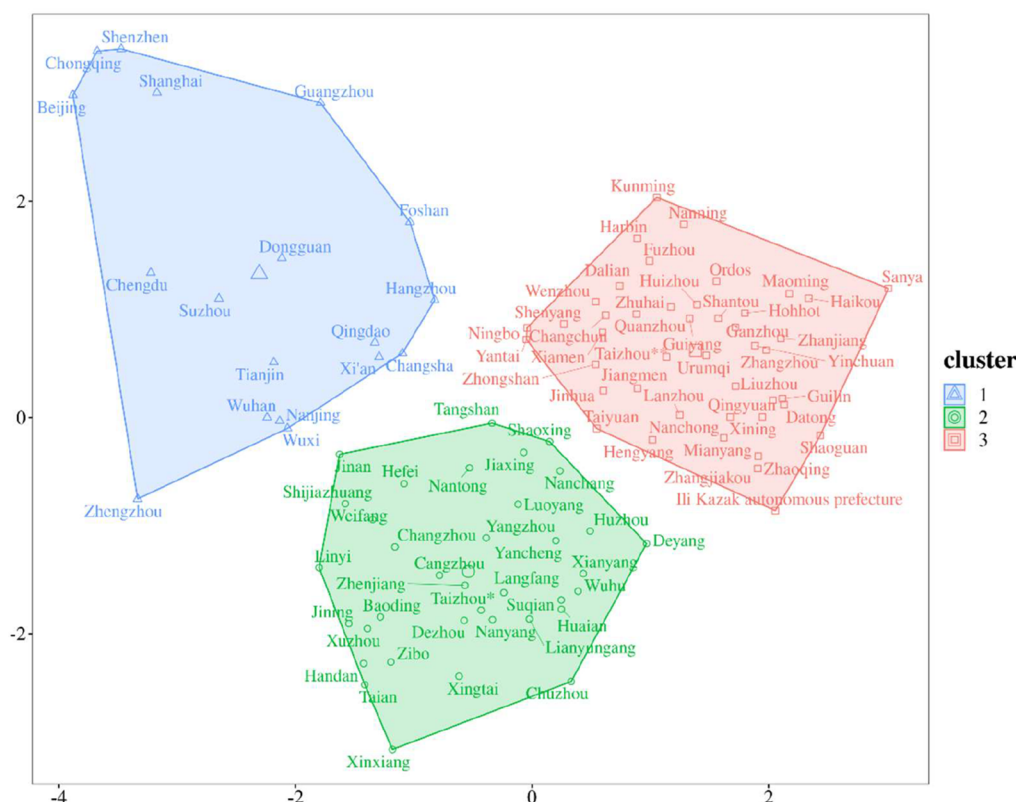


Figure 6. Three clustering categories in 98 Chinese cities. Cluster 1: traffic-intensive cities; Cluster 2: traffic-polluted cities; Cluster 3: traffic-friendly cities. Taizhou*: Taizhou city, Jiangsu province; Taizhou*: Taizhou city, Zhejiang province.

indicators and additional ACS onsets) should prioritize the implementation of emissions reduction strategies.

Traffic-related $PM_{2.5}$ may induce adverse cardiovascular effects and even higher risk levels. The results of the present study illustrated a significantly stronger association between traffic-related $PM_{2.5}$ and ACS onset than previously reported estimation of $PM_{2.5}$,²² partially attributable to its high proportion of ultrafine particles and/or black carbon.^{17,18,49} Previous research proposed possible biological and physiopathological mechanisms for the observed associations.²⁹ Traffic-related $PM_{2.5}$ had been linked to inflammatory reactions, cardiovascular-related depressions, changes and damages, which may contribute to the onset and exacerbation of cardiovascular diseases.^{50–54} A variety of inflammatory cells and mediators (e.g., mononuclear macrophages) contributed to the occurrence and development of atherosclerotic lesions and plaque formation, significantly triggering ACS onset.^{50,55} These adverse effects may be further amplified among susceptible subpopulations. Older adults, who typically exhibit diminished cardiovascular functional reserve and a higher prevalence of pre-existing cardiometabolic conditions, may be particularly vulnerable to the inflammatory and oxidative stress responses triggered by traffic-related $PM_{2.5}$.⁵⁶ In addition, sex-specific differences in hormonal regulation and inflammatory pathways may partly explain the stronger associations observed in females, particularly after menopause when estrogen's cardioprotective effects decline.^{57,58} Collectively, these underlying biological susceptibilities may amplify the triggering effect of traffic-related $PM_{2.5}$ on acute coronary events.

In the current study, reductions of additional ACS onsets were estimated from 2015 to 2018, but increased concentrations and corresponding onsets were observed in some

regions. Air quality in China has significantly improved, particularly with regard to $PM_{2.5}$ emission control. Indeed, as a result of China's efforts to improve air quality, the corresponding additional onsets due to traffic-related exposure also decreased, demonstrating clear health benefits from improved air quality. This is encouraging for further decision-making. Nevertheless, increases or limited improvements in traffic-related $PM_{2.5}$ were still observed in certain regions, resulting in a considerable number of additional onsets. Such regional heterogeneity may be attributable to differences in traffic intensity, vehicle fleet composition—particularly the proportion of diesel-powered heavy-duty vehicles—and the frequency of cross-regional freight transport.^{59,60} Additionally, inconsistent implementation and effectiveness of local traffic emission control measures may contribute to the observed spatiotemporal variations in traffic-related $PM_{2.5}$ concentrations.⁶¹ For example, relatively higher onsets were also observed in some boundary regions, such as the Sichuan–Chongqing–Guizhou transprovincial region, which was partially due to these regions having relatively high traffic densities, particularly diesel-powered heavy-duty vehicles transporting coal and raw materials across provincial borders. It suggested that stricter control measures should be carried out in these key regions, including electrification of interprovincial logistics fleets and the establishment of low-emission freight corridors integrated with renewable energy infrastructure.

Precise emission reduction strategies for various cities should be adopted to combat the adverse effects of traffic-related $PM_{2.5}$ pollution. Cluster analysis provided a framework for policymakers to implement prioritized mitigation strategies. Our study of 98 Chinese cities using cluster analysis identified

three distinct categories. Specifically, cities characterized by lower per capita GDP, moderate population density, and relatively low numbers of motor vehicles and traffic congestion indices yet exhibited exceptionally high levels of traffic-related $PM_{2.5}$ concentrations, leading to significantly increased incidences of ACS and excessive health burdens. Cities classified as cluster 2, such as Zibo, Handan, Xinxiang, and Xuzhou, were marked by concentrated industrial activities and a high reliance on road freight transport.⁶² The local vehicle fleet in these cities contained a larger proportion of diesel-powered heavy-duty vehicles, which were major contributors to traffic-related $PM_{2.5}$ emissions.⁶³ Moreover, compared with megacities, the implementation and enforcement of traffic emission controls, including fleet upgrades and emission supervision, tended to be less stringent or inconsistent, leading to persistently elevated traffic-related $PM_{2.5}$ levels.⁶⁴ These combined factors may explain the coexistence of high pollution levels and increased ACS onset risk in such traffic-polluted cities. To address these issues, targeted strategies are warranted, such as promoting clean-energy vehicles, accelerating the phase-out of older diesel freight trucks, strengthening emission oversight of heavy-duty vehicles, implementing low-emission management along major logistics corridors or within urban zones, and enhancing cross-regional coordinated controls.⁶¹ Conversely, densely populated cities, such as Beijing and Shanghai, characterized by high economic development and vehicle ownership, also demonstrated elevated levels of traffic-related $PM_{2.5}$ emissions and traffic congestion indices. For these traffic-intensive cities, demand-side management and stricter vehicle emission standards are particularly relevant. Measures such as congestion mitigation, optimized traffic flow, and continuously tightening emission regulations could effectively reduce traffic-related $PM_{2.5}$ exposure and associated health risks.^{64,65} This finding aligns with existing research, highlighting that economic development and urbanization levels are key factors influencing motorization rates and $PM_{2.5}$ pollution.⁶⁶ Controlling motor vehicle growth and implementing traffic restrictions are critical pathways to significantly reduce traffic-related $PM_{2.5}$ emissions and safeguard public health.

This study has the following strengths. First, the self-controlled feature of the employed case-crossover design, especially for factors that remained unchanged between case and control days (e.g., individual characters and lifestyles), allows us to adjust for potential confounding. Second, the National Chest Pain Center in China, the largest nationwide network, provides comprehensive coverage and high-quality hospital-based ACS records, offering individual patient information with desensitization to enhance statistical power. Third, the results are the first quantified evidence linking $PM_{2.5}$ and ACS onset in China, providing the unique possibility to investigate associations using large-scale ACS cases and a wide range of traffic-related $PM_{2.5}$. This will help analyze additional ACS onsets, examining potential health benefits under significant declines in $PM_{2.5}$ concentrations in China. Fourth, comparisons of health benefits due to per unit of concentration reduction provide solid evidence that future emission control should focus on traffic-related $PM_{2.5}$, and different detailed suggestions for different city categories are proposed based on the cluster analysis.

There are also some limitations to this analysis. First, we used the source-oriented CMAQ model to generate nationwide county-level concentrations of traffic-related $PM_{2.5}$, which

may introduce an information bias in exposure estimation. Considering no available monitoring data of $PM_{2.5}$ sources, the CMAQ model had become an alternative method and be integrated into epidemiologic analyses.^{67–69} Nevertheless, without considering indoor or outdoor time-activity patterns, exposure misclassification was inevitable. Second, exposure estimation was determined according to hospital addresses registered in the network of the National Chest Pain Center rather than residential addresses or the locations where the event of ACS onset cases occurred, which may likely lead to underestimation, making the reported associations conservative. However, this limitation could be negligible, given that ACS patients always seek medical treatment at the nearest hospital to receive timely treatment due to their characteristics of acute attacks and severe conditions. Third, ACS onset estimation relied on the 5-year average ACS hospitalization rates in Beijing and the county-level population in 2015. Due to limited data availability, the data were drawn from the Beijing Municipal Health Big Data and Policy Research Center, and the average rate for ACS was calculated based on the entire city's registration, which was the best and most publicly available data on ACS and population in China.

Individual-level protective behaviors play a critical role in mitigating associated health risks. The effectiveness of several personal protective measures has been discussed in prior research, such as avoiding commuting in the rush hour, closing windows when driving or living in polluted locations, restricting activities in near-roadway locations, and wearing masks (e.g., N95 or N99) outdoors.^{51,70} Individuals with pre-existing cardiovascular diseases and older adults are especially vulnerable to short-term increases in traffic-related $PM_{2.5}$ exposure and should be prioritized for targeted protective measures. For populations at high ACS risk, integrating air quality information into early intervention and exposure avoidance guidance could help reduce short-term environmental health threats.^{71,72} The results could also be embedded in the policies of the National Chest Pain Center to reinforce the concept that prevention is as important as treatment. Hospitals and doctors in this largest collaborative healthcare network should enhance health education and encourage patients to reduce their exposure to surrounding traffic pollution and follow protective suggestions. By integrating air pollution risk awareness into clinical workflows, healthcare providers could improve timely risk communication, reinforce preventive measures, and promote early intervention among ACS-vulnerable patients.^{73,74} Such institution-based strategies could contribute to a more comprehensive and actionable prevention framework. Notably, the Chinese government should continuously adhere to transportation reform, and substitution to clean and renewable energy should be encouraged in the current fossil fuel-based transport. Tightening vehicle emission standards has been proven to be the most viable approaches, driving a shift from fossil fuels to green energy resources. Besides, efforts should be made to plan decisions or policies in cities for communities and/or special units (e.g., aged-care centers) to balance transportation convenience and traffic-related pollution.

5. CONCLUSIONS

This work elucidates the robust association between traffic-related $PM_{2.5}$ and ACS onset using a nationwide case-crossover analysis. Meanwhile, the health benefits of reduced traffic-related $PM_{2.5}$ exposure on ACS onsets are quantified in China.

Traffic-related PM_{2.5} pollution deserves more concern from the point of view of associations, additional onsets, and avoided benefits. The findings will also support decision-making in fossil fuel management aiming to protect public health. The Chinese government should continuously adhere to transportation reform, and substitution to clean and renewable energy should be promoted in the current fossil fuel-based transportation system. Notably, we also suggest that prioritized emission reduction should be conducted in the identified traffic-polluted cities (cluster 2). We also suggest that collaborative actions from individuals, hospitals and doctors, and governments are appreciated to reduce traffic exposures to prevent ACS onset.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/envhealth.5c00615>.

Supplementary methods regarding guidelines for the diagnosis and treatment of acute coronary syndrome (ACS), estimation of traffic-related PM_{2.5}; results of short-term effects during multiple-day averaging lags, classified by region and season; the estimated additional ACS onsets; descriptive statistics and sensitivity analyses; and average and relative levels of indicators in the cluster analysis (PDF)

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Notes

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